

Title: Graph Neural Network Framework for Financial Transaction Fraud Detection

Abstract

Financial transaction systems generate millions of digital transactions every day, making them attractive targets for fraudulent activities that result in significant financial losses for banks, businesses, and customers. Traditional fraud detection techniques and conventional machine learning models primarily analyze transactions independently and often fail to capture the complex relationships among customers, accounts, merchants, and devices. This project proposes a Graph Neural Network Framework for Financial Transaction Fraud Detection that models financial transactions as interconnected graphs, enabling the learning of relational patterns and hidden fraud networks for accurate fraud detection.

The proposed framework utilizes transaction datasets such as IEEE-CIS Fraud Detection and PaySim to construct graph-based representations where nodes represent entities including customers, accounts, merchants, and devices, while edges represent financial transactions. The system performs data preprocessing, graph construction, feature engineering, graph embedding, and fraud classification using Graph Neural Network architectures such as Graph Convolutional Networks (GCN), GraphSAGE, or Graph Attention Networks (GAT). The framework further generates fraud risk scores and supports explainable predictions through graph-based interpretation techniques to improve model transparency.

The proposed approach can identify suspicious transaction patterns, detect fraud rings, and analyze high-risk financial activities in real time. The performance of the Graph Neural Network model is evaluated using accuracy, precision, recall, F1-score, ROC-AUC score, confusion matrix, and precision-recall metrics. The resulting system aims to provide accurate, scalable, and intelligent financial fraud detection while assisting financial institutions in minimizing fraudulent transactions, reducing operational risks, and improving decision-making through actionable insights.

Datasets

Dataset	URL Link
IEEE-CIS Fraud Detection	https://www.kaggle.com/competitions/ieee-fraud-detection
PaySim - Synthetic Financial Datasets	https://www.kaggle.com/datasets/ealaxi/paysim1
Elliptic Data Set	https://www.kaggle.com/code/smlopezza/elliptic-data-set-eda-graphs-random-forest

Approach

Financial Transaction Datasets



Data Preprocessing



Graph Construction



Feature Engineering



Graph Neural Network (GCN / GraphSAGE / GAT)



Fraud Classification



Risk Score Generation



Explainable AI



Fraud Analytics Dashboard

Project Group Members

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